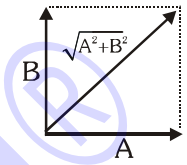
SOLUTION

SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (C)

Sol. $y(x,t) = A\sin\left(kx - \omega t + \frac{\pi}{2}\right) + B\sin(kx - \omega t)$



It is combination of two wave of phase difference $\frac{\pi}{2}$.

$\therefore$ Resultant amplitude = $\sqrt{A^2 + B^2}$

Compare with the wave y (x, t) = D sin(kx–ωt +ϕ)

$\therefore$ $D^2 = A^2 + B^2$

2. Ans. (C)

Sol. A_i positive

$A_r = \frac{V_2 - V_1}{V_2 - V_1} A_i$ $\quad V_2 < V_1$ = negative

3. Ans. (D)

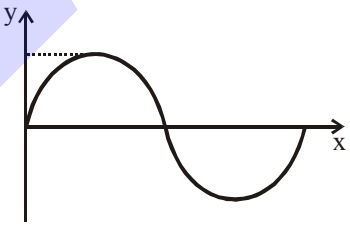
Sol. Wave moving in +ve x direction

At t = 0

At t = 0, x = 0, y = 0

ϕ = 0

y = A sin(kx – ωt)



Wave moving in –ve x direction

y = A sin (ωt + kx + ϕ)

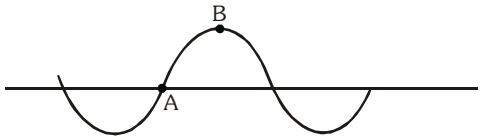
at t = 0, x = 0, y = 0

ϕ = 0

y = 0 A sin(ωt + kx)

4. Ans. (B)

Sol.



Both KE and PE are maximum at y = 0 and minimum at y = A.

V = 0 at point B so KE = 0.

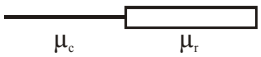
Slope zero at B so PE = 0.

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5. Ans. (C)

Sol.



$A = 1$

A_t – Amplitue of transmitted wave.

$$A_t = \left[\frac{2\sqrt{\mu_\ell}}{\sqrt{\mu_\ell}+\sqrt{\mu_r}}\right]$$

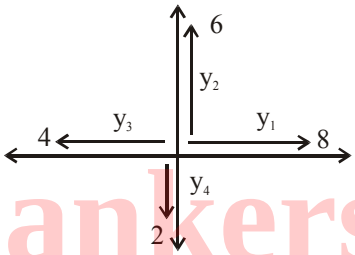
6. Ans. (A)

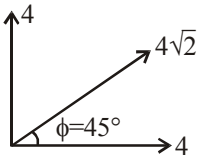
Sol. $y_1 = 8\sin(\omega t + kx)$

$y_2 = 6\sin(\omega t + kx + \pi/2)$

$y_3 = 4\sin(\omega t + kx + \pi)$

$y_4 = 2\sin(\omega t + kx + 3\pi/2)$





$y_5 = 4\sqrt{2}\sin(\omega t - kx + \pi/4)$

$y_1 + y_2 + y_3 + y_4$

$y' = 4\sqrt{2}\sin(\omega t + kx + \pi/4)$

$y_{net} = y' + y_5$

$= 4\sqrt{2}\ (\sin(\omega t - kx + \pi/4) + \sin(\omega t + kx + \pi/4)$

$= 4\sqrt{2}\ [2\sin(\omega t + \pi/4)\cos kx]$

$$= 8\sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)\cos kx$$

7. Ans. (B)

Sol. $\text{T.E.} = \text{KE} + \text{P.E.} = \int\left[\frac{1}{2}(\mu dx)v_P^2 + \frac{1}{2}Tdx\left(\frac{dy}{dx}\right)^2\right] = m^2Ta$

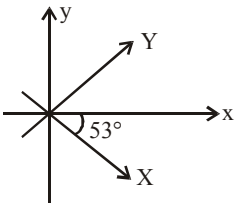
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8. Ans. (C)

Sol. In new cartesian coordinate system X-Y, wave equation can be written as

$\phi = \phi_n \sin(2t - x + \frac{\pi}{3})$



9. Ans. (B)

Sol. The given equation can be written in the form $y = 2 \times 2 \sin\left(\frac{2\pi x}{30}\right)\cos 2\pi\frac{(1440t)}{30}$

$a = 2$ cm, $\lambda = 30$ cm, $v = 1440$ cm/sec; $y_{max} = 4 \sin\frac{2\pi x}{30}$

For $x = 5$, $y_{max} = 4 \sin\frac{2\pi \times 5}{30} = \frac{4\sqrt{3}}{2} = 2\sqrt{3}$ cm

Multiple Correct Answer Type **9 Q. [4 M (–1)]**

10. Ans. (A,B,C,D)

Sol. $y = A \sin(kx - \omega t + \phi)$

at $t = 0$, $x = 0$, $y = 5$ mm

at $t = \frac{1}{24}$, $x = 0$, $y = 0$

also $\lambda = 2m$, $A = 10$ mm

Solve to get answer.

11. Ans. (B, C, D)

12. Ans. (C,D)

Sol. Time taken by point P from one extreme to other is $\frac{T}{2}$

So, $\frac{T}{2} = 0.1s$

In time $t = \frac{T}{4}$, energy flows from node to antinodes.

Velocity $= \sqrt{\frac{T}{\mu}} = 10 m/s$, frequency is 5 Hz

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$\frac{nv}{2\ell} = f \quad \Rightarrow \quad n = 2$

So, the string oscillates with two loops

and x = 1 is a node.

Also x = 0.5 is antinode and at t = 0.05

rate of energy flow through P is zero

13. Ans. (A,B,C,D)

14. Ans. (A, B, C, D)

Sol. At node $\cos(10\pi x) = 0$ & at antinode $\cos(10\pi x) = 1$

& $\omega = 50\ \pi$, $k = 10\pi$ & $v = \frac{\omega}{k}, k = \frac{2\pi}{\lambda}$

15. Ans. (A, C)

Sol. To produce node at x = 0

$y_1 + y_2 = 0 \quad x = 0$

$\Rightarrow A \sin\left(-ut + \frac{\pi}{3}\right) + \frac{1}{2} = 0 \Rightarrow \frac{1}{2} = -A \sin\left(-4t + \frac{\pi}{3}\right)$

Phase diffrance between two wave force fig are π

$\frac{1}{2}$ may be $= A \sin\left(-2x - 4t + \frac{\pi}{3} \pm \pi\right)$

16. Ans. (A, C)

Sol. Let $\lambda_1 = \lambda$

$\lambda_2 = 2\lambda$

Let of wave become 2 time's have frequency will remain same.

17. Ans. (A,B,C,D)

18. Ans. (A,B,D)

Sol. PE and KE both maximum at y = 0. [Simultaneously for two particles in phase]

Amplitude and frequency same for all particles.

There will be phase diffrence in motion of all particles.

Linked Comprehension Type (2 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

19. Ans. (A)

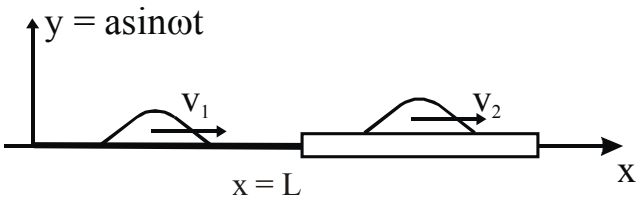
20. Ans. (C)

21. Ans. (B)

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Sol. (19 to 21)



$$Y_i = a\sin\omega\left(t-\frac{x}{v_1}\right)$$

Transmitted at x = L

$$Y_{(x=L)} = a\sin\omega\left(t-\frac{L}{v_1}\right)$$

$$Y_t = a_t\sin\omega\left[t-\frac{L}{v_1}-\left(\frac{x-\ell}{v_2}\right)\right]$$

Reflected at x = L

$$y_{x=2L} = a\sin\omega\left(t-\frac{\ell}{v_1}\right)$$

$$y_{Reff} = a_r\sin\omega\left[t-\frac{\ell}{v_1}+\left(\frac{x-\ell}{v_1}\right)+\pi\right]$$

22. Ans. (D)

Sol. 

A = 3 cm

f = 100 Hz

$$2m = \frac{3\lambda}{2}$$

$$\lambda = \frac{4}{3}m$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi\times 3}{4} = \frac{3\pi}{2}$$

23. Ans. (B)

Sol. $E = (\text{No of loop's})\times\left(\frac{1}{8}\omega^2A_S^2\lambda\right)$

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24. Ans. (C)

Sol. $V = f \times \lambda$ $\qquad \lambda = \frac{4}{3}$

$$\lambda = 100 \times \frac{4}{3} = \frac{400}{3}$$

$$\text{time} = \frac{\lambda}{4V} = \frac{4}{4 \times 3} \times \frac{3}{400} = \frac{1}{400}$$

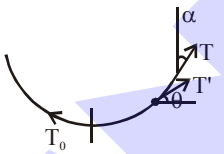
25. Ans. (D)

Sol. $T\cos\alpha = \frac{Mg}{2}$

$T \sin \alpha = T_0$

$T' \cos\alpha = T_0$

$$T'\sin\theta = \frac{M}{L} xg$$



$$\tan\theta = \frac{dy}{dx} = \frac{2x}{L}\cot\alpha$$

$$y = \frac{x^2}{L}\cot\alpha$$

26. Ans. (A)

Sol. $T\cos\alpha = \frac{Mg}{2}$

$T\sin\alpha = T_0$

$$\tan\alpha = \frac{2T_0}{Mg}$$

$$v = \sqrt{\frac{T_0}{m}\ell} = \sqrt{\frac{Mg\ell\tan\alpha}{2m}}$$

SECTION-II

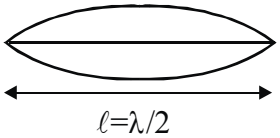
Numerical Answer Type Question **1Q.[3M (0)]**

(upto second decimal place)

27. Ans. 375

Sol. $\rho = \frac{m}{A\ell} \Rightarrow \ell = \frac{m}{\rho A}$ and $\mu = \frac{m}{\ell} \Rightarrow v = \sqrt{\frac{T}{\mu}}$

fundamental frequency



$$f = \frac{v}{\lambda}$$

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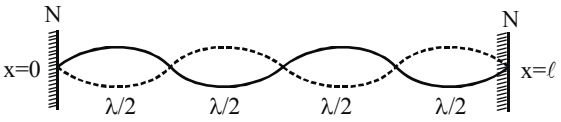
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 1 Q. [4 M (0)]

28. Ans. 9

Sol. We have taken x = 0 as node so that

Amplitude = (A_{max} sink x)

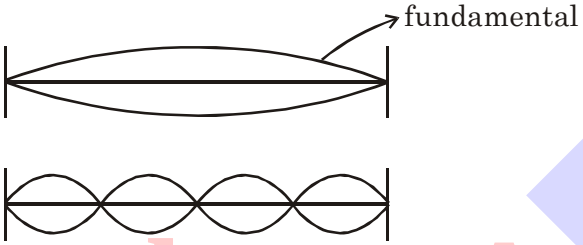


$$K=\frac{2\pi}{\lambda} \text{ where } \lambda=\frac{\ell}{2}\Rightarrow A=A_{max}\sin\left[\frac{2\pi}{\frac{\ell}{2}}\times\frac{\ell}{3}\right]=A\sin\frac{4\pi}{3}=\frac{\sqrt{3}A_{max}}{2}=3$$

Subjective Type 1 Q. [4 M (0)]

29. Ans. 17.5°

Sol.



$$f=\frac{n}{2\ell}\sqrt{\frac{T}{\mu}}$$

$$f^2=\frac{4}{\ell^2}\times\frac{Y\alpha\Delta\theta A}{\rho A}$$

$$f^2=\frac{4}{\ell^2}\times\frac{Y\alpha\Delta\theta}{\rho}$$

$$\Delta\theta=\frac{f^2\ell^2\rho}{4Y\alpha}$$

$$\Delta\theta=2.5^\circ C$$

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